4. Basics of Probabilistic Modelling

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FS26 MTH345: Deep Generative Models: A Statistical Perspective





Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- In this chapter, we formulate Bayesian networks and review two fundamental paradigms in probabilistic modelling: (i) discriminative and (ii) generative models.
- After discussing their limitations in high-dimensional settings, we will transition into traditional dimension reduction methods.
- Recommended reference materials include:
  - Efron (1975), "The Efficiency of Logistic Regression to Normal Discriminant Analysis," *JASA*.
  - Section 14 in Hastie (2009), *The Elements of Statistical Learning*.

Outline

1. Conditional Distribution

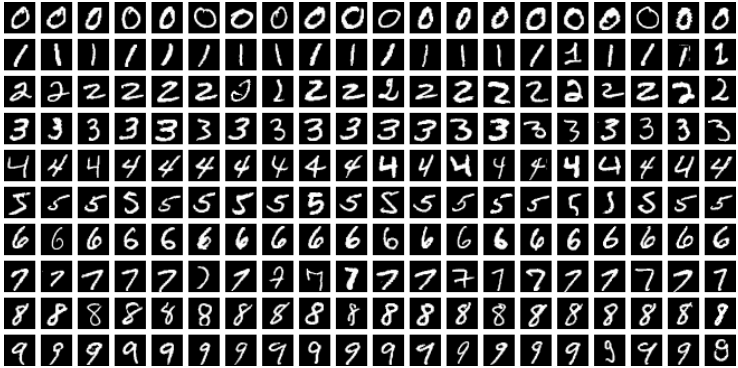
2. Bayesian Network

3. Discriminative vs. Generative Models

4. Curse of Dimensionality

5. Dimensionality Reduction

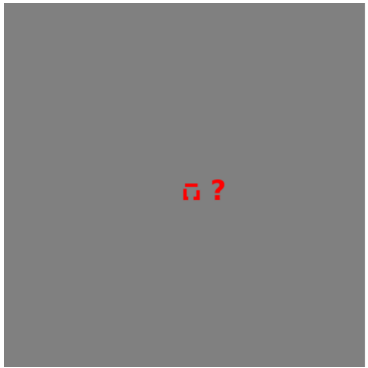
Motivating Example



- Consider images from the Modified National Institute of Standards and Technology (MNIST) database (LeCun, 1998).
- Each image consists of $28 \times 28 = 784$ binary pixels, with values in $\{0, 1\}$.

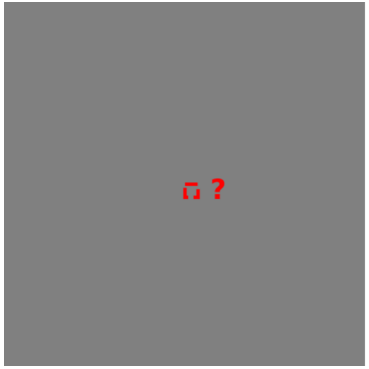
Image source: `https://en.wikipedia.org/wiki/MNIST_database`.

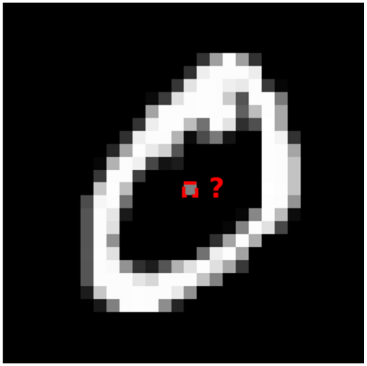
Motivating Example



Outcome (x)	0 (black)	1 (white)
$P(X_{14\times 14} = x)$	48.08%	51.92%

Motivating Example





Outcome (x)	0 (black)	1 (white)
$P(X_{14\times14} = x \mid \boldsymbol{X}_{-14\times14})$	48.08% $\rightarrow$?	51.92% $\rightarrow$?

Conditional Distribution

- In probabilistic modeling, we often seek to understand how the distribution of one random variable changes conditionally on another.
- Recall that conditional probability $P(\cdot \mid B)$ defines a valid probability measure, i.e., it satisfies the three probability axioms.
- Therefore, standard theoretical properties and notions of probability are naturally inherited in the conditional setting.
- **Conditional (Cumulative) Distribution Function**: For random variables X and Y, the function

$$F_{X|Y}(x \mid y) = P(X \leq x \mid Y = y) \tag{1}$$

is called the *conditional cumulative distribution function* (or simply the *conditional distribution function*) of X given $Y = y$, provided that $p_Y(y) > 0$ or $f_Y(y) > 0$.

Conditional Distribution

- **Conditional Probability Mass Function**: For discrete random variables X and Y with joint PMF $p_{X,Y}(x, y)$, the *conditional PMF* of X given $Y = y$ is defined as:

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}, \tag{2}$$

  provided that $p_Y(y) > 0$.

- **Conditional Probability Density Function**: For continuous random variables X and Y with joint PDF $f_{X,Y}(x, y)$, the *conditional PDF* of X given $Y = y$ is defined as:

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \tag{3}$$

  provided that $f_Y(y) > 0$.

- X and Y are independent if and only if $f_{X|Y}(x \mid y) = f_X(x)$ for all x and y.

Conditional Distribution

Exercise 1

Y		x_1	x_2	x_3	x_4	x_5
	y_1	0.01	0.02	0.03	0.1	0.1
	y_2	0.05	0.1	0.05	0.07	0.2
	y_3	0.1	0.05	0.03	0.05	0.04

X

([MML] Exercise 6.1) Consider $p_{X,Y}(x, y)$, whose values are in the table above.

1. Compute $p_X(x)$ and $p_Y(y)$
2. Compute $p_{X|Y}(x \mid Y = y_1)$ and $p_{Y|X}(y \mid X = x_3)$.

Conditional Distribution

Exercise 2

([FCP] Example 6.5a) The joint PDF of X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} \frac{12}{5}x(2-x-y), & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

Compute the conditional PDF of X given that $Y = y$, where $0 < y < 1$.

Conditional Distribution

Exercise 3

Let $(X_1, X_2)^\top$ be a continuous r.v. with joint CDF $F_{X_1,X_2}(x_1, x_2)$. Assume that F_{X_1} is strictly increasing, and that for every x_1, the $F_{X_2|X_1}(x_2 \mid x_1)$ is strictly increasing w.r.t. x_2. Let $U_1, U_2 \overset{\text{i.i.d.}}{\sim} \text{Unif}(0,1)$ and define:

$$Y_1 = F^{-1}_{X_1}(U_1) \quad \text{and} \quad Y_2 = F^{-1}_{X_2|X_1}(U_2 \mid Y_1).$$

Show that the random vector $(Y_1, Y_2)^\top$ has joint CDF F_{X_1,X_2}.

Conditional Distribution



Conditional Distribution

- **Conditional Variance**: The *conditional variance* of X given $Y = y$ is defined as:

$$\mathrm{Var}_{X|Y}[X \mid Y = y] = \mathrm{E}_{X|Y}\left[(X - \mathrm{E}[X \mid Y = y])^2 \mid Y = y\right] \tag{5}$$

where $\mathrm{E}_{X|Y}[g(X) \mid Y = y] = \int_{\mathbb{R}} g(x) f_{X|Y}(x \mid y) dx$ denotes the conditional expectation of $g(X)$ given $Y = y$.

Exercise 4

Law of Total Variance: $\mathrm{Var}_X[X] = \mathrm{E}_Y[\mathrm{Var}_{X|Y}[X \mid Y]] + \mathrm{Var}_Y[\mathrm{E}_{X|Y}[X \mid Y]]$

Conditional Distribution



Conditional Independence

- **Conditional Independence:** Two random variables X and Y are said to be *conditionally independent* given Z if, for any three intervals of real numbers I_X, I_Y, and I_Z with $P(Z \in I_Z) > 0$,

$$P(X \in I_X, Y \in I_Y \mid Z \in I_Z) = P(X \in I_X \mid Z \in I_Z) P(Y \in I_Y \mid Z \in I_Z). \quad (6)$$

  That is, the events $X^{-1}(I_X)$ and $Y^{-1}(I_Y)$ are conditionally independent given $Z^{-1}(I_Z)$.
- We write $X \perp\!\!\!\perp Y \mid Z$ to denote that X is independent of Y given Z.

Conditional Independence

Exercise 5

1. Let $X \sim \text{Ber}(0.5)$ and $Y \sim \text{Ber}(0.5)$ be independent. Let $Z = X + Y$. Check whether $X \perp\!\!\!\perp Y \mid Z$ holds.
2. Let $Z \sim \text{Ber}(0.5)$ and define $X = Y = Z$. Check whether $X \perp\!\!\!\perp Y \mid Z$ holds.

Conditional Independence

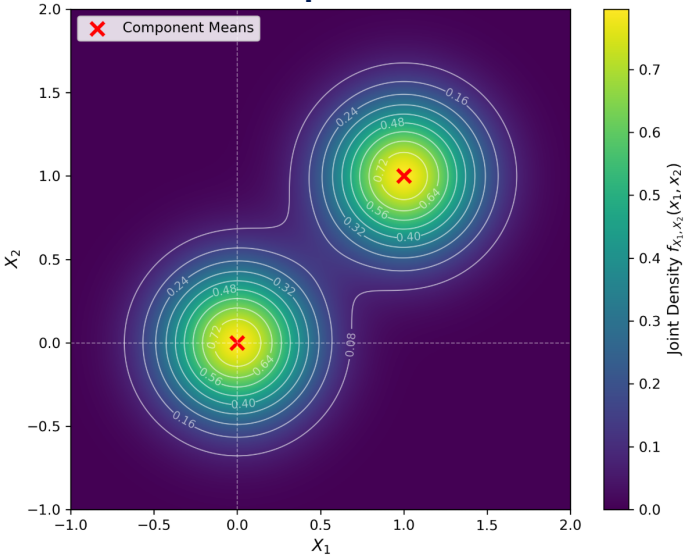
- **Conditional Covariance**: The *conditional covariance* between X and Y given $Z = z$ is defined as:

$$\mathrm{Cov}[X, Y \mid Z = z] = \mathrm{E}\left[\big(X - \mathrm{E}[X \mid Z = z]\big)\big(Y - \mathrm{E}[Y \mid Z = z]\big) \;\Big|\; Z = z\right]. \tag{7}$$

- As with (unconditional) covariance, if $X \perp\!\!\!\perp Y \mid Z$, then $\mathrm{Cov}[X, Y \mid Z = z] = 0$ for all z. However, the converse does not hold in general.
- **Law of Total Covariance**:

$$\mathrm{Cov}[X, Y] = \mathrm{E}[\mathrm{Cov}[X, Y \mid Z]] + \mathrm{Cov}[\mathrm{E}[X \mid Z], \mathrm{E}[Y \mid Z]]. \tag{8}$$

Conditional Independence



$$\begin{cases} Y \sim \text{Ber}(0.5), \\ \boldsymbol{X} \mid Y = 0 \sim \mathcal{N}\left((0,0)^\top, 0.1\boldsymbol{I}_2\right), \\ \boldsymbol{X} \mid Y = 1 \sim \mathcal{N}\left((1,1)^\top, 0.1\boldsymbol{I}_2\right). \end{cases}$$

Conditional Independence

Exercise 6

Let $(X_1, X_2, Y)^\top$ be generated from the following hierarchical model:

$$Y \sim \text{Ber}(p) \quad \text{and} \quad \boldsymbol{X} \mid Y = y \sim \mathcal{N}(\boldsymbol{\mu}_y, \sigma^2 \boldsymbol{I}_2), \tag{9}$$

where $0 < p < 1$, and $\boldsymbol{\mu}_0 \neq \boldsymbol{\mu}_1$.

1. Show that $\text{Cov}[X_1, X_2] = p(1-p)(\mu_{1,1} - \mu_{0,1})(\mu_{1,2} - \mu_{0,2})$.
2. Show analytically that $f_{X_1, X_2 \mid Y}(x_1, x_2 \mid y) = f_{X_1 \mid Y}(x_1 \mid y) f_{X_2 \mid Y}(x_2 \mid y)$.

Conditional Independence



Outline

1. Conditional Distribution

2. Bayesian Network

3. Discriminative vs. Generative Models

4. Curse of Dimensionality

5. Dimensionality Reduction

Bayesian Network

- The *chain rule* can factor any joint PDF $f_{\boldsymbol{X}}(\boldsymbol{x})$ of a random vector $\boldsymbol{X} = (X_1, \ldots, X_d)^\top$ into a product of conditional densities.
- For example, when $d = 3$,

$$\begin{aligned} f_{\boldsymbol{X}}(\boldsymbol{x}) &= f_{X_1}(x_1) f_{X_2|X_1}(x_2 \mid x_1) f_{X_3|X_1,X_2}(x_3 \mid x_1, x_2) \\ &= \cdots \\ &= f_{X_3}(x_3) f_{X_2|X_3}(x_2 \mid x_3) f_{X_1|X_2,X_3}(x_1 \mid x_2, x_3). \end{aligned} \tag{10}$$

- By repeatedly applying the definition of conditional probability:

$$f(x_1, \ldots, x_d) = f(x_1) \prod_{i=2}^{d} f(x_i \mid x_1, \ldots, x_{i-1}).$$

Q: How do we model the dependence within $\boldsymbol{X}$ or the joint PDF $f(x_1, \ldots, x_d)$?

Bayesian Network

1. **Full Model**: Without any domain knowledge, we can model all the conditional distributions,

$$f(x_i \mid x_1, \ldots, x_{i-1}). \tag{11}$$

- It can express any joint distribution, but the model size becomes large.

2. **Degenerate Model**: When $(X_2, \ldots, X_d)$ are known deterministic functions of X_1, modelling $f(X_1)$ alone is sufficient.

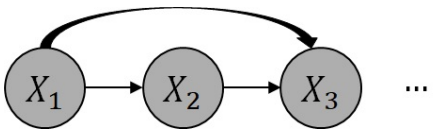
- The model size is small and even invariant to the data dimension. However, the model capacity is significantly reduced.

Bayesian Network

- *Bayesian networks* are probabilistic graphical models that represent conditional dependencies using directed acyclic graph structures.
- Again, we can express the joint PDF $f(x_1, \ldots, x_d)$ as

$$f(x_1)f(x_2 \mid x_1) \cdots f(x_d \mid x_1, x_2, \ldots, x_{d-1}). \tag{12}$$

Many dependency structures can be represented by a directed graph with d nodes $X_1, \ldots, X_d$ and up to $d(d-1)/2$ directed edges.

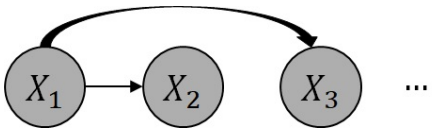


Bayesian Network

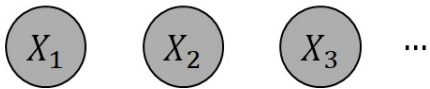
- For example, **1. Full Model** corresponds to a graph utilizing all nodes and all possible directed edges.
- In general, joint PDFs specified by a Bayesian network factors into the product of the conditional densities of each variable given its parents, denoted by $\mathrm{Pa}(X_i)$:

$$f(x_1, \ldots, x_d) = \prod_{i=1}^{d} f(x_i \mid \mathrm{Pa}(x_i)).$$

Bayesian Network



2. **Degenerate Model**: This model is a special case of the above graph where $f(x_1, \ldots, x_d) = f(x_1)f(x_2 \mid x_1) \cdots f(x_d \mid x_1)$.



3. **Fully Independent Model**: This model assumes mutual independence among all variables, such that $f(x_1, \ldots, x_d) = f(x_1) \cdots f(x_d)$.

Bayesian Network: Latent Variable Model



4. **Latent Variable Model**: There are latent factors $\boldsymbol{Z} := (Z_1, \ldots, Z_r)^\top$, typically consisting of independent components, that are mixed to generate data, e.g., $\boldsymbol{X} = W\boldsymbol{Z}$ in Independent Component Analysis (Hyvärinen and Oja, 2000).

- The age, light source location, and camera angle are examples of latent factors.

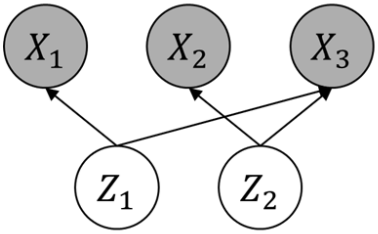
The top and bottom images are from the Extended Yale-B (Georghiades et al., 2001) and Multi-pie (Gross et al., 2010) datasets, respectively.

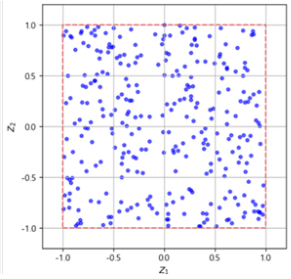
Bayesian Network: Latent Variable Model

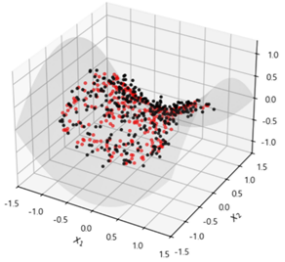
Exercise 7

Let $\boldsymbol{Z} = (Z_1, \ldots, Z_r)^\top$ with $Z_j \overset{\text{i.i.d.}}{\sim} \text{Unif}(0,1)$ be a latent vector, and define the observed random vector $\boldsymbol{X} = W\boldsymbol{Z}$ where $W \in \mathbb{R}^{r \times r}$ is an invertible matrix. Express the marginal PDF $f_{\boldsymbol{X}}(\boldsymbol{x})$.

Bayesian Network: Latent Variable Model







- For example, suppose Z_1 and $Z_2 \overset{\text{i.i.d.}}{\sim} \text{Unif}(-1, 1)$ and $\boldsymbol{X}$ is:

$$X_1 = Z_1 + \epsilon_1, \quad X_2 = Z_2 + \epsilon_2, \quad X_3 = (Z_1^2 - Z_2^2)/2 + \epsilon_3 \tag{13}$$

  where ϵ represents white noise, e.g., $\mathcal{N}(0, 1)$.
- Here, Z_1 and Z_2 are *parent nodes* with no incoming edges, while X_1, X_2, and X_3 are *child nodes* with ones.

Bayesian Network: Latent Variable Model

Exercise 8

Let $\boldsymbol{Z} = (Z_1, Z_2)^\top$ with $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \text{Unif}(-1, 1)$ be a latent vector, and let $\boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2, \epsilon_3)^\top$ with $\epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ be observation noise independent of $\boldsymbol{Z}$. Suppose the observed vector $\boldsymbol{X} = (X_1, X_2, X_3)^\top$ is generated according to:

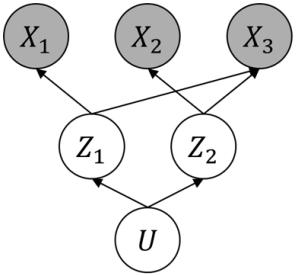
$$X_1 = Z_1 + \epsilon_1, \quad X_2 = Z_2 + \epsilon_2, \quad X_3 = (Z_1^2 - Z_2^2)/2 + \epsilon_3. \tag{14}$$

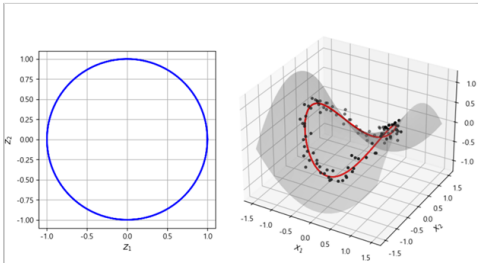
Express the marginal PDF $f_{\boldsymbol{X}}(\boldsymbol{x})$.

Bayesian Network: Latent Variable Model



Bayesian Network: Latent Variable Model





- Now, suppose $U \sim \text{Unif}(0,1)$ and $\boldsymbol{Z} \mid U = u \sim \mathcal{N}\left((\cos 2\pi u, \sin 2\pi u)^\top, \sigma^2 \boldsymbol{I}_2\right)$, where $\sigma > 0$ is small.

Bayesian Network: Latent Variable Model

- The joint PDF of $(U, Z_1, Z_2, X_1, X_2, X_3)^\top$ factorizes according to the network structure:

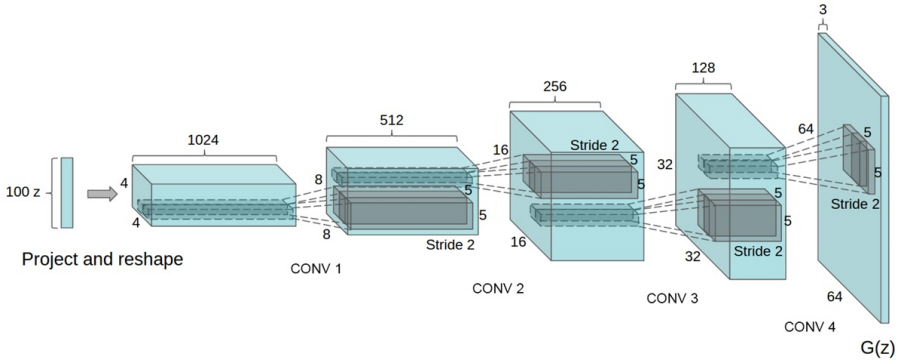
$$
\begin{aligned}
& f(u, z_1, z_2, x_1, x_2, x_3) \\
&\quad = f(u) f(z_1, z_2 \mid u) f(x_1, x_2, x_3 \mid u, z_1, z_2) \\
&\quad = f(u) f(z_1, z_2 \mid u) f(x_1, x_2, x_3 \mid z_1, z_2) \\
&\quad = f(u) f(z_1 \mid u) f(z_2 \mid u) f(x_1 \mid z_1) f(x_2 \mid z_2) f(x_3 \mid z_1, z_2) \\
&\quad = I(0 \le u \le 1) \left(\frac{1}{\sigma} \phi \left(\frac{z_1 - \cos 2\pi u}{\sigma} \right) \cdot \frac{1}{\sigma} \phi \left(\frac{z_2 - \sin 2\pi u}{\sigma} \right) \right) \\
&\qquad \times \left(\phi(x_1 - z_1) \cdot \phi(x_2 - z_2) \cdot \phi \left(x_3 - \frac{z_1^2 - z_2^2}{2} \right) \right).
\end{aligned}
\tag{15}
$$

Bayesian Network: Latent Variable Model

- The marginal PDF $f_{\boldsymbol{X}}(\boldsymbol{x})$ is obtained by marginalizing over $(u, z_1, z_2)^\top \in (0,1) \times \mathbb{R}^2$:

$$
\begin{aligned}
f_{\boldsymbol{X}}(\boldsymbol{x}) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{1} f(u, z_1, z_2, x_1, x_2, x_3)\, du\, dz_1\, dz_2 \\
&= \frac{1}{(2\pi)^{5/2}\sigma^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{1} \exp\left(-\frac{1}{2}\left(\frac{(z_1-\cos 2\pi u)^2}{\sigma^2} + \frac{(z_2-\sin 2\pi u)^2}{\sigma^2}\right.\right. \\
&\qquad \left.\left. + (x_1-z_1)^2 + (x_2-z_2)^2 + \left(x_3 - \frac{z_1^2 - z_2^2}{2}\right)^2\right)\right) du\, dz_1\, dz_2.
\end{aligned}
$$

Bayesian Network: Latent Variable Model



- Deep generative models are predominantly based on latent variable models.
- The conditional distributions $f(\boldsymbol{x}|\boldsymbol{z})$ can be modeled as parametric distributions, e.g., $\mathcal{N}(\boldsymbol{\mu}_\theta(\boldsymbol{z}), \boldsymbol{\Sigma}_\theta(\boldsymbol{z}))$ in Variational Autoencoders (Kingma and Welling, 2014).

An example neural network architecture for deep generative model from Radford (2015).

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1. Conditional Distribution

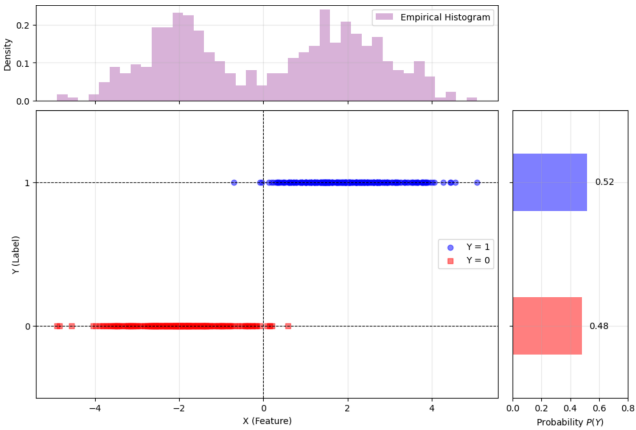
2. Bayesian Network

3. Discriminative vs. Generative Models

4. Curse of Dimensionality

5. Dimensionality Reduction

Discriminative vs. Generative Models



- Given data, there are multiple ways to describe the dependency structure.
- For example, let X be an explanatory variable and Y be a binary response.
- **Q**: How can we construct a probabilistic model for $(X, Y)^\top$ and learn $p(y \mid x)$?[a]

[a] This task is referred to as *binary classification*.

Discriminative vs. Generative Models



(a) Discriminative Model (b) Generative Model

- We may consider two approaches:
  1. **Discriminative Model**: Models only the conditional distribution $p(y|x)$.
  2. **Generative Model**: Models the joint distribution

$$f(x, y) = p(y)f(x \mid y), \tag{16}$$

typically by modelling $p(y)$ and $f(x|y)$ separately.

Discriminative Model



(a) Discriminative Model (b) Generative Model

- **Discriminative Model**: We consider *logistic regression*. First, we may consider:

$$X \sim f(x) \quad \text{and} \quad Y \mid X = x \sim \text{Ber}(\mu_\theta(x)), \tag{17}$$

where $f(x)$ does not depend on the parameter θ, and $\mu_\theta(x)$ is a parameterized conditional success probability designed to model the dependence of Y on X.

Discriminative Model

- In this case, for a given x, the conditional log-PMF is expressed as:

$$\begin{aligned}\log p_\theta(y \mid x) &= \log \left(\mu_\theta(x)^y (1 - \mu_\theta(x))^{1-y} \right) \\ &= y \log \frac{\mu_\theta(x)}{1 - \mu_\theta(x)} + \log(1 - \mu_\theta(x)).\end{aligned} \tag{18}$$

- Note that $\log p_\theta(y \mid x)$ is a linear function of y with the coefficient:

$$\eta_\theta(x) = \log \frac{\mu_\theta(x)}{1 - \mu_\theta(x)},^1 \tag{19}$$

which is the *logit* (or *log-odds*) of success given x.

[1] It is formally referred to as the *natural parameter* in the context of the *exponential family*.

Discriminative Model

- Here, letting $\mathrm{sig}(u) := 1/(1+\exp(-u))$ denote the *sigmoid function* (or *logistic function*), we have:

$$\mu_\theta(x) = \frac{1}{1+\exp(-\eta_\theta(x))} = \mathrm{sig}(\eta_\theta(x)).^2 \tag{20}$$

- For the choice of $\eta_\theta(\cdot)$, we consider the *logistic model* where the logit $\eta_\theta(x)$ is a linear function of x:

$$\left\{ \eta_\theta(x) = \beta_0 + \beta_1 x \;\middle|\; \theta = (\beta_0, \beta_1)^\top \in \mathbb{R}^2 \right\}. \tag{21}$$

- Finding the MLE within this model class leads to *logistic regression*.

[2] In a similar manner, this formulation can be extended to multi-class classification using the *softmax function*.

Discriminative Model

Exercise 9

Let $\boldsymbol{X} \sim f(\boldsymbol{x})$ and $\boldsymbol{Y} \mid \boldsymbol{X} = \boldsymbol{x} \sim \text{Multin}(1, \text{Softmax}(\boldsymbol{\eta}_{\boldsymbol{\theta}}(\boldsymbol{x})))$,[a] where $\boldsymbol{\eta}_{\boldsymbol{\theta}}(\boldsymbol{x}) := (\boldsymbol{\beta}_1^\top \boldsymbol{x}, \ldots, \boldsymbol{\beta}_K^\top \boldsymbol{x})^\top$ and the Softmax denotes the *softmax function*:

$$\text{Softmax}(\boldsymbol{u})_k := \frac{\exp(u_k)}{\sum_{j=1}^K \exp(u_j)}, \quad k \in \{1, \ldots, K\}.$$

Show that:

$$\log p_{\boldsymbol{\theta}}(\boldsymbol{y} \mid \boldsymbol{x}) = \boldsymbol{\eta}_{\boldsymbol{\theta}}(\boldsymbol{x})^\top \boldsymbol{y} - \log \left(\sum_{k=1}^K \exp(\boldsymbol{\beta}_k^\top \boldsymbol{x}) \right). \tag{22}$$

[a] In general, $\boldsymbol{x}$ includes 1 to model intercept, i.e., $\boldsymbol{x} = (1, x_1, \ldots, x_d)^\top$.

Discriminative Model



Discriminative Model

- Let's formulate the likelihood maximization. The (joint) log-likelihood function can be expressed as:

$$
\begin{aligned}
\ell(\theta) &= \sum_{i=1}^{n} \log f_\theta(x_i, y_i) \\
&= \sum_{i=1}^{n} \log f(x_i) + \sum_{i=1}^{n} \log p_\theta(y_i \mid x_i) \\
&= \sum_{i=1}^{n} \log f(x_i) + \sum_{i=1}^{n} \left(\log \left(\frac{\mu_\theta(x_i)}{1 - \mu_\theta(x_i)} \right) y_i + \log \left(1 - \mu_\theta(x_i)\right) \right) \\
&= \sum_{i=1}^{n} \log f(x_i) + \sum_{i=1}^{n} \left(\eta_\theta(x_i) y_i + \log \left(1 - \operatorname{sig}(\eta_\theta(x_i))\right) \right).
\end{aligned}
\tag{23}
$$

Discriminative Model

Exercise 10

Let $X \sim f(x)$, $Y \mid X = x \sim \text{Ber}(\mu_\theta(x))$, and $\{(x_i, y_i)\}_{i=1}^n$ be n observations.

1. Show that $\text{sig}'(u) = \text{sig}(u)(1 - \text{sig}(u))$.
2. Let $\eta_\theta(x) = \text{sig}^{-1}(\mu_\theta(x))$ denote the logit function. Show that:

$$\nabla_\theta \ell(\theta) = \sum_{i=1}^{n} \left(y_i - \text{sig}(\eta_\theta(x_i))\right) \nabla_\theta \eta_\theta(x_i). \tag{24}$$

Discriminative Model

3. Assuming $\eta_\theta(x) = \beta_0 + \beta_1 x$ where $\theta = (\beta_0, \beta_1)^\top$, show that the *Hessian matrix* can be expressed as:

$$\nabla_\theta^2 \ell(\theta) = -\sum_{i=1}^{n} \text{sig}(\eta_\theta(x_i)) \left(1 - \text{sig}(\eta_\theta(x_i))\right) \nabla_\theta \eta_\theta(x_i) \nabla_\theta \eta_\theta(x_i)^\top. \quad (25)$$

Discriminative Model

- For the linear logistic model $\eta_\theta(x) = \beta_0 + \beta_1 x$, the gradient of the logit is:

$$\nabla_\theta \eta_\theta(x_i) = (1, x_i)^\top. \tag{26}$$

  Consequently, the Hessian $\nabla^2_\theta \ell(\theta)$ is strictly negative definite, provided that the sample variance of the covariates satisfies $S^2_X > 0$.

- Thus, $\ell(\theta)$ is strictly concave. If a solution to the *score equation*,

$$\nabla_\theta \ell(\theta) = 0, \tag{27}$$

  exists, it is the unique maximizer of the log-likelihood.[3]

- We can apply Newton's method, an iterative optimization algorithm, to numerically approximate the MLE.

[3] The solution exists if and only if the data are not completely or quasi-completely separable by a hyperplane.

Discriminative Model

Exercise 11

Let $X \sim f(x)$, $Y \mid X = x \sim \text{Ber}(\mu_\theta(x))$, $\{(x_i, y_i)\}_{i=1}^n$ be n observations, and $\eta_\theta(x) = \text{sig}^{-1}(\mu_\theta(x)) = \beta_0 + \beta_1 x$ with $\theta = (\beta_0, \beta_1)^\top$. Show that $\nabla_\theta \ell(\theta) = 0$ is equivalent to:

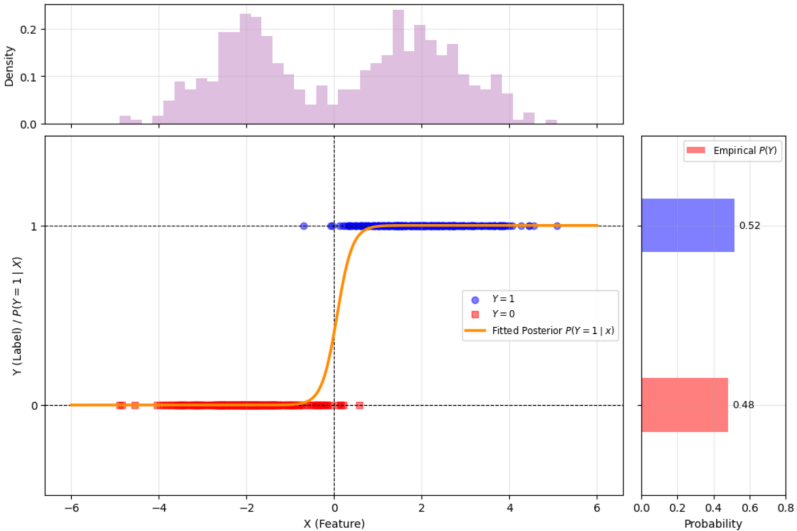
$$\begin{cases} \dfrac{1}{n}\displaystyle\sum_{i=1}^n \hat{p}_i = \dfrac{1}{n}\sum_{i=1}^n y_i, \\ \dfrac{1}{n}\displaystyle\sum_{i=1}^n (x_i - \bar{x})(\hat{p}_i - \bar{\hat{p}}) = \dfrac{1}{n}\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \end{cases} \tag{28}$$

where $\hat{p}_i := \mu_\theta(x_i) = P_\theta(Y = 1 \mid X = x_i)$ and $\bar{\hat{p}} = n^{-1}\sum_{i=1}^n \hat{p}_i$.

Discriminative Model



Discriminative Model



Generative Model



(a) Discriminative Model (b) Generative Model

- **Generative Model**: We consider *linear discriminant analysis (LDA)*. First, we model the joint distribution directly:

$$Y \sim \mathrm{Ber}(p) \quad \text{and} \quad X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2), \tag{29}$$

where $p \in (0, 1)$ is the class prior, and the conditional distribution of X given Y is modeled as a class-specific Gaussian with a shared variance σ^2.

Generative Model

- Let $\theta = (p, \mu_0, \mu_1, \sigma^2)^\top$ be the parameter vector. Then, the joint probability can be expressed as:

$$f_\theta(x, y) = \underbrace{\left(p^y(1-p)^{1-y}\right)}_{=p_\theta(y)} \underbrace{\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_y)^2}{2\sigma^2}\right)\right)}_{=f_\theta(x|y)}. \tag{30}$$

- Consequently, by Bayes' rule, the posterior probabilities are given by:

$$P_\theta(Y = 1 \mid X = x) = \frac{f_\theta(x \mid y = 1)p_\theta(y = 1)}{f_\theta(x \mid y = 1)p_\theta(y = 1) + f_\theta(x \mid y = 0)p_\theta(y = 0)} \tag{31}$$

and

$$P_\theta(Y = 0 \mid X = x) = \frac{f_\theta(x \mid y = 0)p_\theta(y = 0)}{f_\theta(x \mid y = 1)p_\theta(y = 1) + f_\theta(x \mid y = 0)p_\theta(y = 0)}. \tag{32}$$

Generative Model

Exercise 12

Let $Y \sim \text{Ber}(p)$ and $X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2)$ where $\theta = (p, \mu_0, \mu_1, \sigma^2)^\top$. Show that the logit can be expressed as:

$$\eta_\theta(x) = \left(\log \frac{p}{1-p} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2} \right) + \left(\frac{\mu_1 - \mu_0}{\sigma^2} \right) x \tag{33}$$

Generative Model

- Consequently, the decision boundary of this model, defined as the set of points where $\eta_\theta(x) = 0$, is strictly *linear* with respect to x:

$$\beta_0 + \beta_1 x = 0, \tag{34}$$

where we define the mapped parameters as:

$$\beta_0 := \log \frac{p}{1-p} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2} \quad \text{and} \quad \beta_1 := \frac{\mu_1 - \mu_0}{\sigma^2}. \tag{35}$$

- This demonstrates that the LDA naturally induces a logistic regression structure. We can thus use its MLE to calculate the predicted posterior probabilities.
- Conversely, the LDA can represent any (non-trivial) logistic model. When $\beta_1 \neq 0$,

$$(p, \mu_0, \mu_1, \sigma^2)^\top = \left(0.5, -\frac{\beta_0}{\beta_1} - \frac{\beta_1}{2}, -\frac{\beta_0}{\beta_1} + \frac{\beta_1}{2}, 1\right)^\top \tag{36}$$

recovers the corresponding logistic model with $(\beta_0, \beta_1)^\top$.

Generative Model

- Let's formulate the likelihood maximization. The log-likelihood function is given by:

$$\begin{aligned}\ell(\theta) &= \sum_{i=1}^{n} \log\left(p^{y_i}(1-p)^{1-y_i}\right) + \sum_{i=1}^{n} \log\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu_{y_i})^2}{2\sigma^2}\right)\right) \\ &= \sum_{i=1}^{n} \left(y_i \log p + (1-y_i)\log(1-p) - \frac{1}{2}\log(2\pi\sigma^2) - \frac{(x_i-\mu_{y_i})^2}{2\sigma^2}\right).\end{aligned} \tag{37}$$

- Unlike logistic regression, $\ell(\theta)$ can be maximized analytically in closed form:

$$\hat{p}_n^{\mathsf{MLE}} = \frac{1}{n}\sum_{i=1}^{n} y_i, \quad \hat{\mu}_{0,n}^{\mathsf{MLE}} = \frac{\sum_{i=1}^{n}(1-y_i)x_i}{\sum_{i=1}^{n}(1-y_i)}, \quad \hat{\mu}_{1,n}^{\mathsf{MLE}} = \frac{\sum_{i=1}^{n} y_i x_i}{\sum_{i=1}^{n} y_i},$$
$$\text{and} \quad \widehat{\sigma}_n^{2,\mathsf{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu}_{y_i,n})^2. \tag{38}$$

Generative Model

Exercise 13

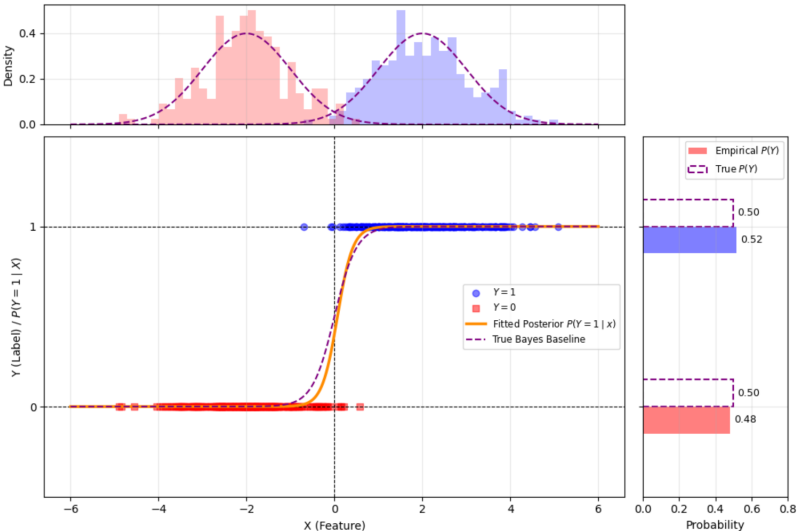
Let $Y \sim \text{Ber}(p)$ and $X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2)$, where $\theta = (p, \mu_0, \mu_1, \sigma^2)^\top$. Given n i.i.d. observations $\{(x_i, y_i)\}_{i=1}^n$ with $0 < \sum_{i=1}^n y_i < n$, show that the maximum likelihood estimator $\hat{\theta}_n^{\text{MLE}} = (\hat{p}_n^{\text{MLE}}, \hat{\mu}_{0,n}^{\text{MLE}}, \hat{\mu}_{1,n}^{\text{MLE}}, \widehat{\sigma}_n^{2,\text{MLE}})^\top$ is given by:

$$\hat{p}_n^{\text{MLE}} = \frac{1}{n}\sum_{i=1}^n y_i, \quad \hat{\mu}_{0,n}^{\text{MLE}} = \frac{\sum_{i=1}^n (1-y_i)x_i}{\sum_{i=1}^n (1-y_i)}, \quad \hat{\mu}_{1,n}^{\text{MLE}} = \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n y_i},$$
$$\text{and} \quad \widehat{\sigma}_n^{2,\text{MLE}} = \frac{1}{n}\sum_{i=1}^n (x_i - \hat{\mu}_{y_i,n}^{\text{MLE}})^2. \tag{39}$$

Generative Model



Generative Model



Generative Model: Synthetic Data Generation

- Given the LDA generative model $f_\theta(x, y) = p_\theta(y) f_\theta(x \mid y)$, we can *generate* synthetic data as follows:
  1. Draw a new label $y_{\text{new}} \sim \text{Ber}(p)$.
  2. Generate a uniform realization $u \sim \text{Unif}(0, 1)$.
  3. Calculate x_{new}:

$$x_{\text{new}} = \mu_{y_{\text{new}}} + \sigma \Phi^{-1}(u). \tag{40}$$

- More generally, for any class-conditional distribution with a known CDF $F_{X|Y=y}(x)$, we can draw $u \sim \text{Unif}(0, 1)$ and compute:

$$x_{\text{new}} = F^{-1}_{X|Y=y_{\text{new}}}(u). \tag{41}$$

Discriminative vs. Generative Models



(a) Discriminative Model (b) Generative Model

- **Q**: What would be the pros and cons of the two approaches?
  (a) **Logistic Regression**: Directly models the posterior probability:

$$P_\theta(Y = 1 \mid X = x) = \mathrm{sig}(\beta_0 + \beta_1 x). \tag{42}$$

  (b) **Linear Discriminant Analysis**: Models the entire data-generating process:

$$f_\theta(x, y) = p_\theta(y) f_\theta(x \mid y), \tag{43}$$

  where $Y \sim \mathrm{Ber}(p)$, $X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2)$, and $\theta = (p, \mu_0, \mu_1, \sigma^2)^\top$.

Discriminative vs. Generative Models

- In the previous toy data, the true data-generating mechanism was as follows:

$$Y \sim \mathrm{Ber}(p) \quad \text{and} \quad X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2) \tag{44}$$

with $(p, \mu_0, \mu_1, \sigma^2)^\top = (0.5, -2, 2, 1)^\top$, implying $(\beta_0, \beta_1)^\top = (0, 4)^\top$. Both models correctly specify the true linear logit structure.

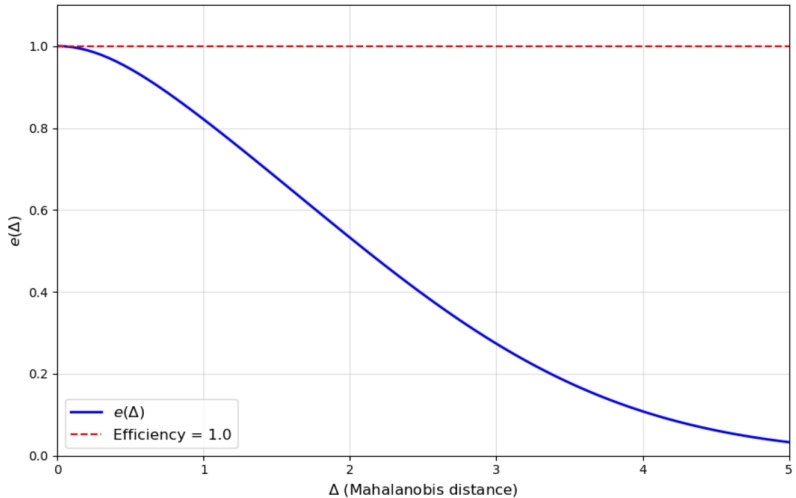
- When $p = 0.5$, the relative asymptotic efficiency for β_1 is expressed as (Efron, 1975):

$$e(\Delta) := \lim_{n \to \infty} \frac{\mathrm{Var}(\hat{\beta}_{1,\mathrm{LDA}})}{\mathrm{Var}(\hat{\beta}_{1,\mathrm{LR}})} = \frac{4 + \Delta^2}{2} \int_{-\infty}^{\infty} z^2 \frac{\phi\left(z - \frac{\Delta}{2}\right) \phi\left(z + \frac{\Delta}{2}\right)}{\phi\left(z - \frac{\Delta}{2}\right) + \phi\left(z + \frac{\Delta}{2}\right)} dz, \tag{45}$$

where $\Delta = \frac{|\mu_1 - \mu_0|}{\sigma}$ represents the *Mahalanobis distance*.

- In our specific setup ($\Delta = 4$), the efficiency for the slope is $e(4) \approx 0.195$.

Discriminative vs. Generative Models



Discriminative vs. Generative Models

- Next, suppose that the true underlying data-generating mechanism follows:

$$Y \sim \mathrm{Ber}(0.5), \quad X \mid Y = 0 \sim -2 + t(2), \quad \text{and} \quad X \mid Y = 1 \sim 2 + t(2). \tag{46}$$

  Here, $t(2)$ denotes a Student's t-distribution with 2 degrees of freedom, whose PDF is given by:

$$f(x) = \frac{1}{2\sqrt{2}} \left(1 + \frac{x^2}{2}\right)^{-3/2}. \tag{47}$$

  Due to this structural form, it exhibits a much higher probability of observing extreme values (heavy tails) than a Gaussian distribution.

- We then investigate the consequences of naively applying both logistic regression and LDA, completely ignoring this model misspecification.

Discriminative vs. Generative Models

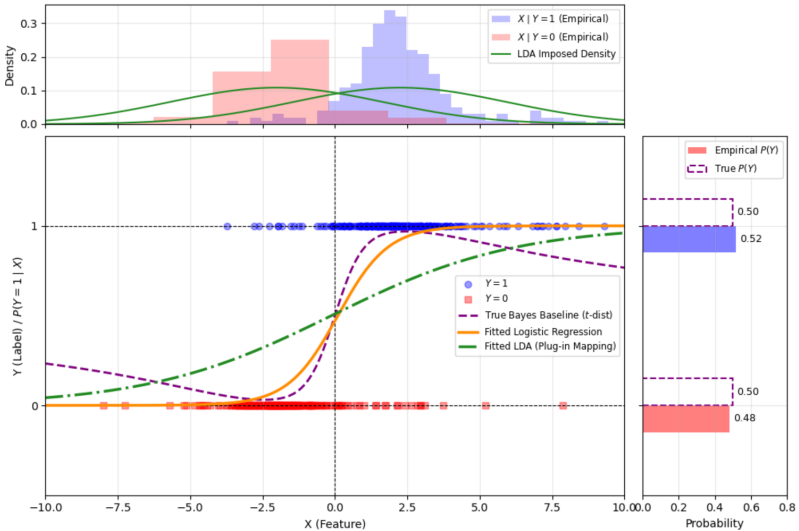


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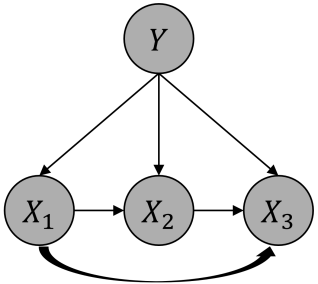
2. Bayesian Network

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4. Curse of Dimensionality

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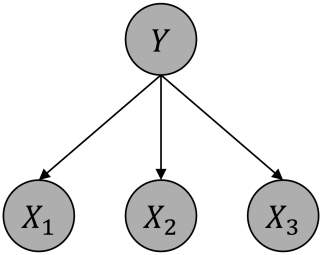
Multivariate Generative Model



- Multivariate data often exhibits complex dependency structures.
- The generative model requires learning $f_\theta(\boldsymbol{x} \mid y)$. For example, when $d = 3$:

$$\begin{aligned} f_\theta(\boldsymbol{x}, y) &= p_\theta(y) f_\theta(\boldsymbol{x} \mid y) \\ &= p_\theta(y) \left(f_\theta(x_1 \mid y) f_\theta(x_2 \mid y, x_1) f_\theta(x_3 \mid y, x_1, x_2) \right). \end{aligned} \tag{48}$$

Multivariate Generative Model: Naive Bayes



- *Naive Bayes* is a multivariate generative model assuming complete conditional independence:

$$f_\theta(\boldsymbol{x} \mid y) = \prod_{j=1}^{d} f_\theta(x_j \mid y). \tag{49}$$

Multivariate Generative Model: Naive Bayes

- The assumption of conditional independence simplifies the logit structure:

$$\begin{aligned}\eta_\theta(\boldsymbol{x}) &= \log \frac{P_\theta(Y=1 \mid \boldsymbol{X}=\boldsymbol{x})}{P_\theta(Y=0 \mid \boldsymbol{X}=\boldsymbol{x})} \\ &= \log \left(\frac{f_\theta(\boldsymbol{x} \mid y=1) p_\theta(y=1)}{f_\theta(\boldsymbol{x} \mid y=0) p_\theta(y=0)} \right) \\ &= \sum_{j=1}^{d} \log \frac{f_\theta(x_j \mid y=1)}{f_\theta(x_j \mid y=0)} + \log \frac{p_\theta(y=1)}{p_\theta(y=0)}.\end{aligned} \tag{50}$$

- The problem of learning $\log \frac{f_\theta(\boldsymbol{x}|y=1)}{f_\theta(\boldsymbol{x}|y=0)}$ reduces to learning d univariate logits.
- For example, if $X_j \mid Y=y \sim \mathcal{N}(\mu_{yj}, \sigma_j^2)$, the logit becomes linear:

$$\eta_\theta(\boldsymbol{x}) = \sum_{j=1}^{d} \left(\frac{\mu_{1j} - \mu_{0j}}{\sigma_j^2} x_j + \frac{\mu_{0j}^2 - \mu_{1j}^2}{2\sigma_j^2} \right) + \log \frac{p_\theta(y=1)}{p_\theta(y=0)}. \tag{51}$$

Multivariate Generative Model: Multivariate LDA

- Unlike Naive Bayes, multivariate LDA models the full dependency structure:

$$\boldsymbol{X} \mid Y = y \sim \mathcal{N}(\boldsymbol{\mu}_y, \Sigma). \tag{52}$$

- The covariance matrix Σ captures dependencies between features and is shared across all classes.
- The conditional PDF is given by:

$$f_\theta(\boldsymbol{x} \mid y) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_y)^T \Sigma^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_y)\right). \tag{53}$$

where $\theta = (\boldsymbol{\mu}_0^\top, \boldsymbol{\mu}_1^\top, \text{vec}(\Sigma))^\top$ is the parameter set containing $O(d^2)$ parameters.

Multivariate Generative Model: Multivariate LDA

Exercise 14

Let $Y \sim \text{Ber}(p)$ and $\boldsymbol{X} \mid Y = y \sim \mathcal{N}(\boldsymbol{\mu}_y, \Sigma)$. Show that the logit can be expressed as:

$$\eta_\theta(\boldsymbol{x}) = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \Sigma^{-1} \left(\boldsymbol{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0}{2} \right) + \log \frac{p}{1-p}. \tag{54}$$

Multivariate Generative Model: Multivariate LDA



Multivariate Generative Model: Multivariate LDA

- The logit,

$$\eta_\theta(\boldsymbol{x}) = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \Sigma^{-1} \left(\boldsymbol{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0}{2} \right) + \log \frac{p}{1-p}, \tag{55}$$

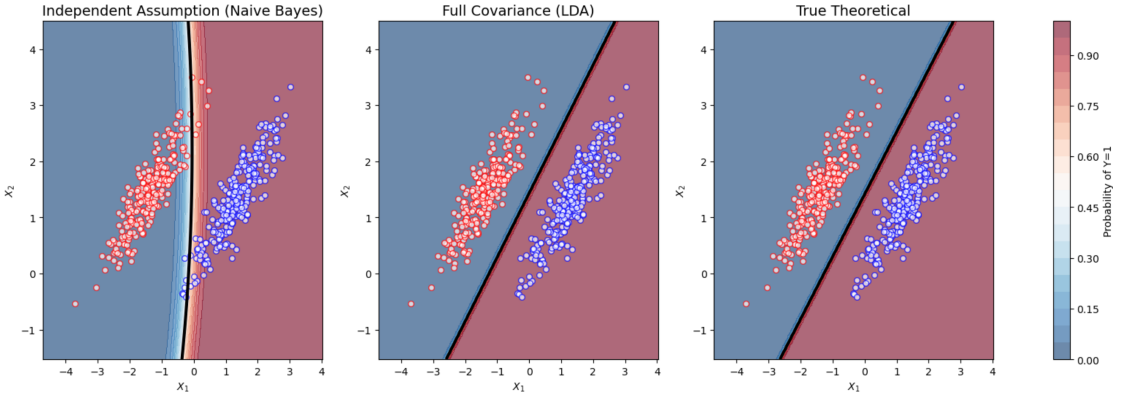
  remains linear. However, Σ^{-1} explicitly accounts for cross-feature dependencies when defining the weight vector $\boldsymbol{w} = \Sigma^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$.

- The MLE is given by:

$$\hat{p}_n^{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n y_i, \quad \hat{\boldsymbol{\mu}}_{0,n}^{\text{MLE}} = \frac{\sum_{i=1}^n (1-y_i)\boldsymbol{x}_i}{\sum_{i=1}^n (1-y_i)}, \quad \hat{\boldsymbol{\mu}}_{1,n}^{\text{MLE}} = \frac{\sum_{i=1}^n y_i \boldsymbol{x}_i}{\sum_{i=1}^n y_i},$$
$$\text{and} \quad \widehat{\Sigma}_n^{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n (\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_{y_i,n}^{\text{MLE}})(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_{y_i,n}^{\text{MLE}})^\top. \tag{56}$$

- When $\Sigma = \sigma^2 \mathrm{I}_d$, multivariate LDA reduces to Gaussian naive Bayes.

Multivariate Generative Model



Curse of Dimensionality

- As d grows much faster than the sample size n ($d/n \to \infty$), multivariate LDA performs no better than random guessing (Bickel and Levina, 2004).
- When $P(Y = 1) = 0.5$, the plug-in classification rule based on multivariate LDA is formulated as:

$$\hat{\delta}_{\text{LDA}}(\boldsymbol{X}) = I\left((\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0)^T \hat{\boldsymbol{\Sigma}}^{-1}\left(\boldsymbol{X} - \frac{\hat{\boldsymbol{\mu}}_1 + \hat{\boldsymbol{\mu}}_0}{2}\right) > 0\right) \tag{57}$$

  where $\hat{\boldsymbol{\mu}}_y$ and $\hat{\boldsymbol{\Sigma}}$ are the MLEs.
- Note that a random guessing classifier defined as:

$$\delta_{\text{Random}}(\boldsymbol{X}) = I(U < 0.5), \quad \text{where } U \sim \text{Unif}(0, 1), \tag{58}$$

  yields a misclassification error rate of $P_\theta(\delta_{\text{Random}}(\boldsymbol{X}) \neq Y) = 0.5$.

Curse of Dimensionality

- The misclassification error rate of multivariate LDA satisfies:

$$P_\theta(\hat{\delta}_{\text{LDA}}(\boldsymbol{X}) \neq Y) \to 0.5 \tag{59}$$

  when $d/n \to \infty$, even if the true covariance matrix Σ is well-conditioned with bounded eigenvalues (e.g., $\Sigma = I_d$) (Bickel and Levina, 2004).
- The root cause of this curse of dimensionality is the excessive number of parameters to estimate.
- When $d \gg n$, the sample covariance matrix $\hat{\Sigma}$ fails to estimate the $O(d^2)$ elements of Σ. Specifically, $\hat{\Sigma}$ becomes singular, making it challenging to compute $\hat{\Sigma}^{-1}$.

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4. Curse of Dimensionality

5. Dimensionality Reduction

Linear Discriminant Analysis

- *Dimension reduction* is the process of identifying low-dimensional features that capture the essential information in the data and transforming the data accordingly.
- For example, the LDA classification decision,

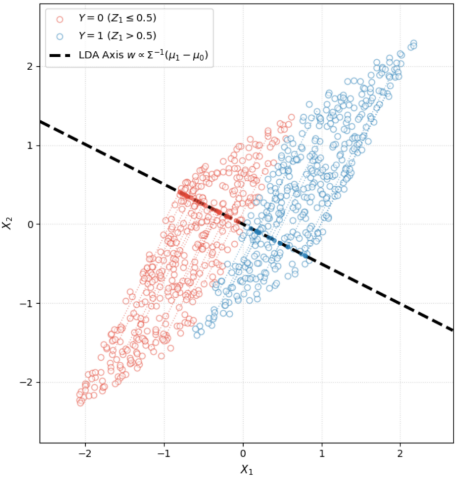
$$\hat{\delta}_{\text{LDA}}(\boldsymbol{X}) = I\left((\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0)^T \hat{\boldsymbol{\Sigma}}^{-1}\left(\boldsymbol{X} - \frac{\hat{\boldsymbol{\mu}}_1 + \hat{\boldsymbol{\mu}}_0}{2}\right) > 0\right), \tag{60}$$

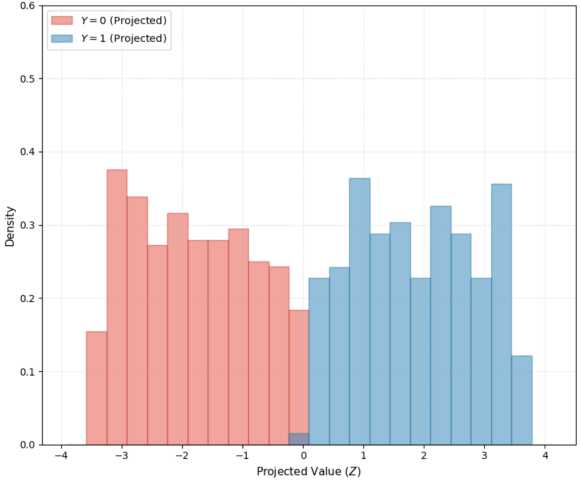
  is based on the scalar projection:

$$\underbrace{\left(\hat{\boldsymbol{\Sigma}}^{-1}(\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0)\right)^T}_{=:\boldsymbol{w}^\top}\left(\boldsymbol{X} - \frac{\hat{\boldsymbol{\mu}}_1 + \hat{\boldsymbol{\mu}}_0}{2}\right). \tag{61}$$

- In this context, LDA can be viewed as a supervised dimension reduction method that utilizes the class label Y.

Linear Discriminant Analysis





Principal Component Analysis

- While LDA effectively identifies the axis that discriminates between the two classes, labeled data are typically scarce in practice.
- Principal Component Analysis (PCA) is a popular unsupervised dimension reduction method that does not utilize class labels.
- PCA first finds the direction that preserves the variability in the data as much as possible, formulated as:

$$\begin{aligned} \max_{\boldsymbol{w}_1} \quad & \widehat{\mathrm{Var}}[\boldsymbol{w}_1^\top \boldsymbol{X}] = \boldsymbol{w}_1^\top \hat{\Sigma} \boldsymbol{w}_1 \\ \text{s.t.} \quad & \boldsymbol{w}_1^\top \boldsymbol{w}_1 = 1, \end{aligned} \tag{62}$$

where $\hat{\Sigma} = (n-1)^{-1} \sum_{i=1}^{n} (\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^\top$ denotes the sample covariance matrix.

Principal Component Analysis

Exercise 15

Let $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$ be n observations. Consider the Lagrangian function

$$\mathcal{L}(\boldsymbol{w}_1, \beta) = \boldsymbol{w}_1^\top \hat{\Sigma} \boldsymbol{w}_1 + \beta \left(1 - \boldsymbol{w}_1^\top \boldsymbol{w}_1\right), \tag{63}$$

where $\hat{\Sigma}$ denotes the sample covariance matrix.

In this problem, you may use the spectral theorem without proof: any real symmetric positive semi-definite matrix A can be decomposed as $A = \sum_{j=1}^{d} \lambda_j \boldsymbol{e}_j \boldsymbol{e}_j^\top$, where the eigenvalues satisfy $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$, and the corresponding eigenvectors $\{\boldsymbol{e}_j\}_{j=1}^{d}$ form an orthonormal basis for $\mathbb{R}^d$.

Principal Component Analysis

1. Show that any stationary point $(\boldsymbol{w}_1, \beta)$ of $\mathcal{L}(\boldsymbol{w}_1, \beta)$ satisfies:

$$\hat{\Sigma}\boldsymbol{w}_1 = \beta\boldsymbol{w}_1 \quad \text{and} \quad \|\boldsymbol{w}_1\|_2 = 1. \tag{64}$$

Principal Component Analysis

2. Since the objective is continuous and $\{\boldsymbol{w} \in \mathbb{R}^d : \|\boldsymbol{w}\|_2 = 1\}$ is compact, a global maximizer exists and must be a stationary point. Show that:

$$\max_{\|\boldsymbol{w}_1\|_2=1} \boldsymbol{w}_1^\top \hat{\Sigma} \boldsymbol{w}_1 = \lambda_1, \tag{65}$$

and that this maximum is achieved at $\boldsymbol{w}_1 = \boldsymbol{e}_1$.

Principal Component Analysis

- Consequently, having obtained the first $k-1$ principal axes $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{k-1}$, the k-th principal axis is sequentially found by maximizing the variance orthogonal to them:

$$\begin{aligned} \max_{\boldsymbol{w}_k} \quad & \boldsymbol{w}_k^\top \hat{\Sigma} \boldsymbol{w}_k \\ \text{s.t.} \quad & \boldsymbol{w}_k^\top \boldsymbol{w}_k = 1, \\ & \boldsymbol{w}_k^\top \boldsymbol{w}_j = 0 \quad \text{for all } j < k. \end{aligned} \tag{66}$$

- The principal axes correspond to the eigenvectors of $\hat{\Sigma}$, where the associated eigenvalues λ_j directly quantify the explained variance along each component:

$$\hat{\Sigma} \boldsymbol{w}_j = \lambda_j \boldsymbol{w}_j \quad \text{where} \quad \lambda_j = \widehat{\text{Var}}[\boldsymbol{w}_j^\top \boldsymbol{X}]. \tag{67}$$

- The total sample variance is preserved under this orthogonal transformation:

$$\sum_{j=1}^{d} \widehat{\text{Var}}[X_j] = \text{tr}(\hat{\Sigma}) = \sum_{j=1}^{d} \lambda_j = \sum_{j=1}^{d} \widehat{\text{Var}}[\boldsymbol{w}_j^\top \boldsymbol{X}]. \tag{68}$$

Principal Component Analysis

Exercise 16

Suppose the first $k-1$ principal axes $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{k-1}$ have been determined as the orthonormal eigenvectors $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_{k-1}$ corresponding to the $(k-1)$-largest eigenvalues. Consider the Lagrangian function:

$$\mathcal{L}(\boldsymbol{w}_k, \beta, \gamma) = \boldsymbol{w}_k^\top \hat{\Sigma} \boldsymbol{w}_k + \beta \left(1 - \boldsymbol{w}_k^\top \boldsymbol{w}_k\right) - 2 \sum_{j=1}^{k-1} \gamma_j \boldsymbol{w}_k^\top \boldsymbol{e}_j, \tag{69}$$

where $\hat{\Sigma}$ denotes the sample covariance matrix.

In this problem, you may use the spectral theorem without proof: any real symmetric positive semi-definite matrix A can be decomposed as $A = \sum_{j=1}^{d} \lambda_j \boldsymbol{e}_j \boldsymbol{e}_j^\top$, where the eigenvalues satisfy $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$, and the corresponding eigenvectors $\{\boldsymbol{e}_j\}_{j=1}^{d}$ form an orthonormal basis for $\mathbb{R}^d$.

Principal Component Analysis

1. Show that any stationary point $(\boldsymbol{w}_k, \beta, \boldsymbol{\gamma})$ of $\mathcal{L}(\boldsymbol{w}_k, \beta, \boldsymbol{\gamma})$ satisfies:

$$\hat{\Sigma}\boldsymbol{w}_k = \beta \boldsymbol{w}_k, \quad \|\boldsymbol{w}_k\|_2 = 1, \quad \text{and} \quad \boldsymbol{w}_k^\top \boldsymbol{e}_j = 0 \quad (\forall j < k). \tag{70}$$

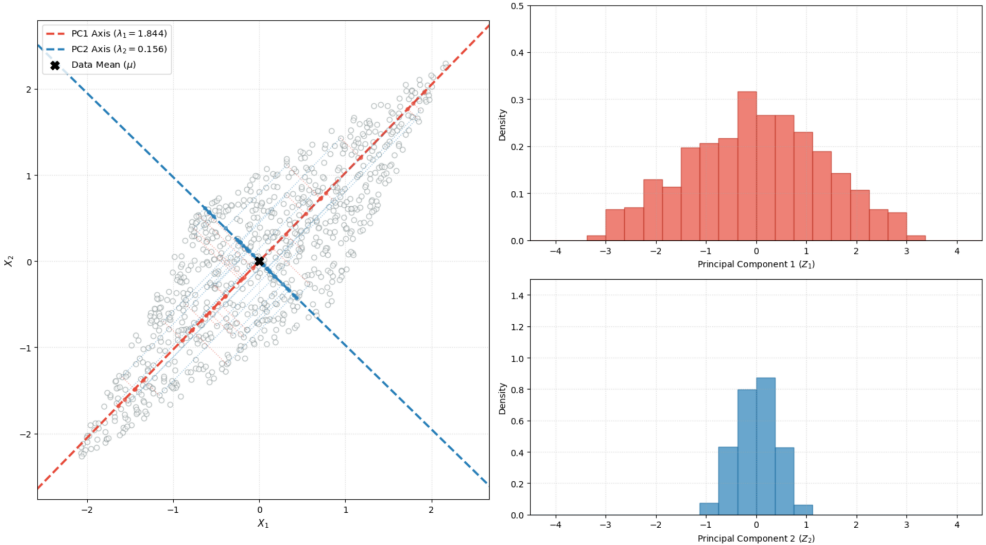
Principal Component Analysis

2. Since the objective is continuous and $\{\boldsymbol{w} \in \mathbb{R}^d : \|\boldsymbol{w}\|_2 = 1,\ \boldsymbol{w}^\top \boldsymbol{e}_j = 0 \text{ for all } j < k\}$ is compact, a global maximizer exists and must be a stationary point. Show that:

$$\max_{\substack{\|\boldsymbol{w}_k\|_2=1 \\ \boldsymbol{w}_k^\top \boldsymbol{e}_j=0\ (\forall j<k)}} \boldsymbol{w}_k^\top \hat{\boldsymbol{\Sigma}} \boldsymbol{w}_k = \lambda_k, \tag{71}$$

and that this maximum is achieved at $\boldsymbol{w}_k = \boldsymbol{e}_k$.

Principal Component Analysis



Principal Component Analysis

- The principal axes can be viewed as optimal linear dimensionality reduction.
- Consider the following mirrored encoding-decoding mapping:

$$\boldsymbol{X} \xrightarrow{\text{Embedding}} W^{\top}(\boldsymbol{X} - \bar{\boldsymbol{x}}) \xrightarrow{\text{Reconstruction}} \bar{\boldsymbol{x}} + WW^{\top}(\boldsymbol{X} - \bar{\boldsymbol{x}}), \tag{72}$$

  where $W = [\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_k] \in \mathbb{R}^{d \times k}$ is the projection matrix.
- The *reconstruction error* can be expressed as:

$$n^{-1} \sum_{i=1}^{n} \left\| \boldsymbol{x}_i - \left(\bar{\boldsymbol{x}} + WW^{T}(\boldsymbol{x}_i - \bar{\boldsymbol{x}}) \right) \right\|^2. \tag{73}$$

Principal Component Analysis

Exercise 17

Consider the following mirrored encoding-decoding mapping:

$$\boldsymbol{x} \xrightarrow{\text{Embedding}} W^\top(\boldsymbol{x}-\bar{\boldsymbol{x}}) \xrightarrow{\text{Reconstruction}} \bar{\boldsymbol{x}} + WW^\top(\boldsymbol{x}-\bar{\boldsymbol{x}}), \tag{74}$$

where $W = [\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_k] \in \mathbb{R}^{d\times k}$ is the projection matrix. Show that if $W^\top W = I_k$,

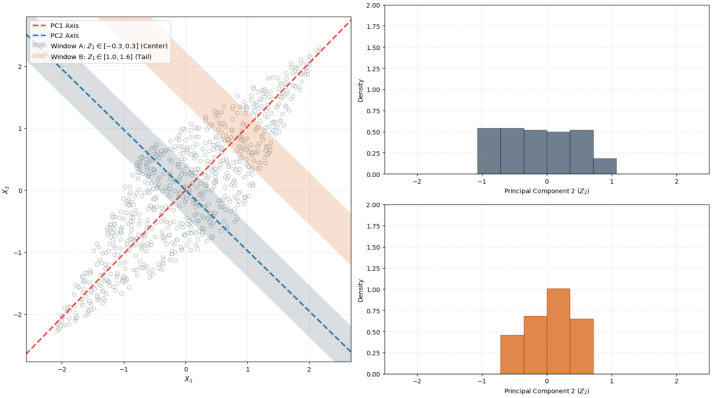
$$(n-1)^{-1}\sum_{i=1}^{n}\left\|\boldsymbol{x}_i - \left(\bar{\boldsymbol{x}} + WW^\top(\boldsymbol{x}_i-\bar{\boldsymbol{x}})\right)\right\|^2 = \operatorname{tr}(\hat{\Sigma}) - \sum_{j=1}^{k}\widehat{\operatorname{Var}}[\boldsymbol{w}_j^\top \boldsymbol{X}], \tag{75}$$

where $\hat{\Sigma}$ denotes the sample covariance matrix.

Principal Component Analysis

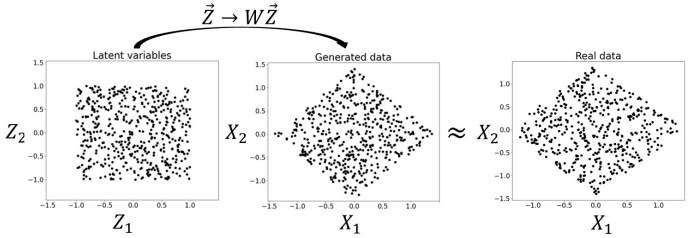


Independent Component Analysis



- Although PCA identifies orthogonal components, orthogonality or uncorrelatedness alone does not imply statistical independence.

Independent Component Analysis



- Independent Component Analysis (ICA) (Hyvärinen and Oja, 2000) identifies latent sources and utilizes them to achieve statistically independent dimension reduction.
- ICA assumes a linear generative structure: $\boldsymbol{X} = W\boldsymbol{Z}$, where $W \in \mathbb{R}^{d \times k}$ and $W^\top W = I_k$ and $\boldsymbol{Z} \sim \text{Unif}(0,1)^k$.
- Assuming all underlying sources are mutually independent, the latent variables can be explicitly inferred via $\boldsymbol{Z} = W^\top \boldsymbol{X}$.

Independent Component Analysis

- There are several criteria for learning this linear generative structure, including kurtosis maximization, mutual information minimization, and maximum likelihood estimation.
- Focusing on the likelihood-maximization approach,[4] the change of variables formula yields:

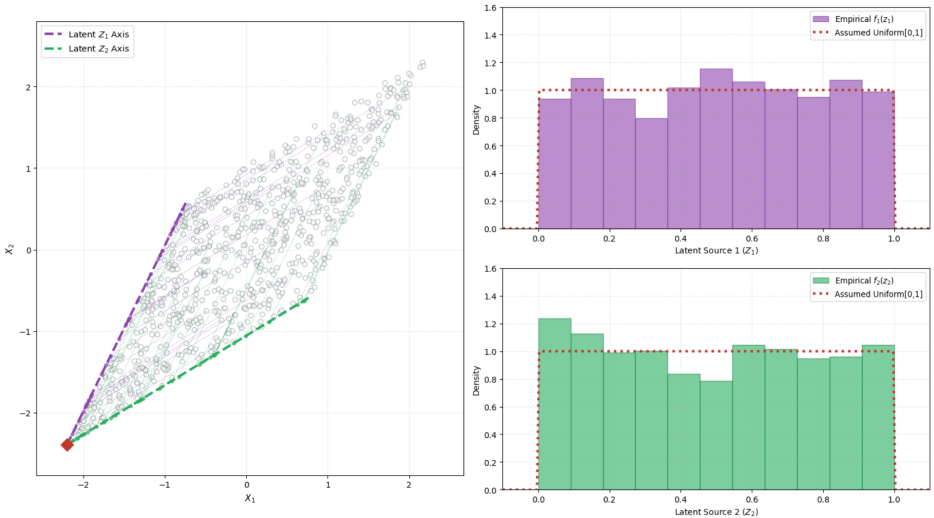
$$f_\theta(\boldsymbol{x}) = f_{\boldsymbol{Z}}(\boldsymbol{z} = W^\top \boldsymbol{x}) \cdot |\det W^T|. \tag{76}$$

- Since W is orthogonal ($|\det W^T| = 1$), the log-likelihood $l_n(\theta)$ simplifies via strict statistical independence to:

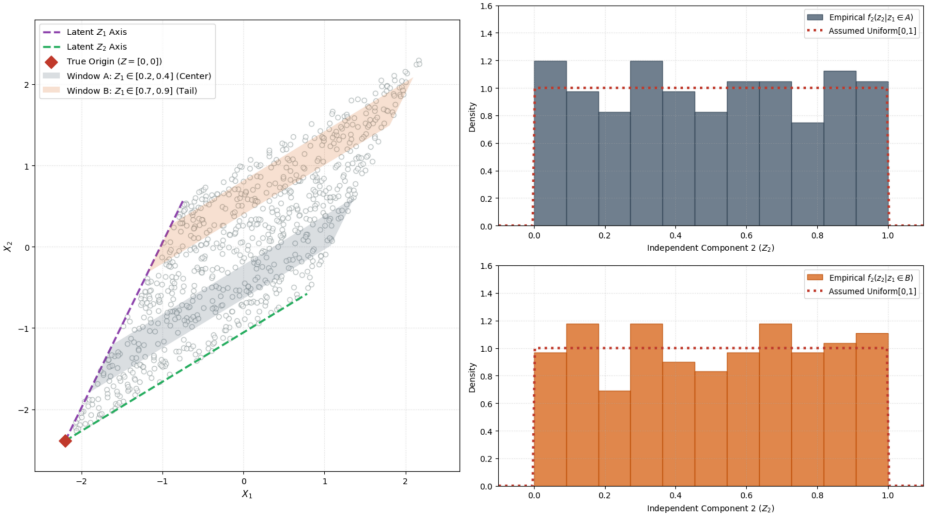
$$l_n(\theta) = \sum_{i=1}^{n} \log f_{\boldsymbol{Z}}(W^\top \boldsymbol{x}_i) = \sum_{i=1}^{n} \sum_{j=1}^{d} \log f_{Z_j}\left((W^\top \boldsymbol{x}_i)_j\right) \tag{77}$$

[4] Although ICA is generally applicable when $k < d$, for simplicity, we assume $k = d$ so that $W^\top = W^{-1}$.

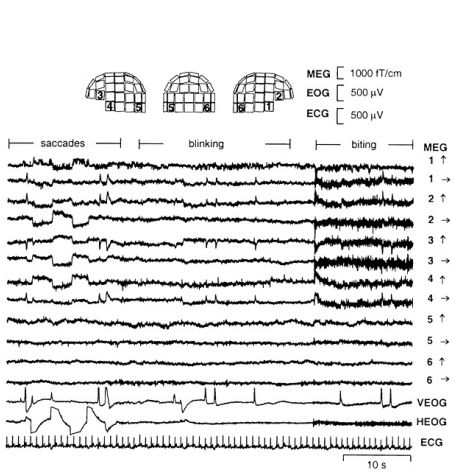
Independent Component Analysis

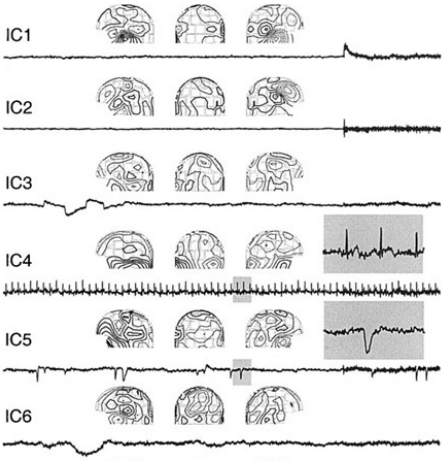


Independent Component Analysis



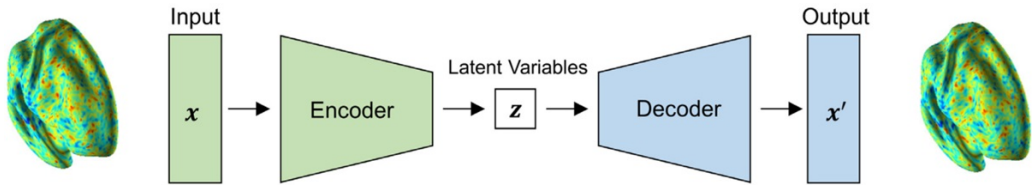
Independent Component Analysis





Figures are edited from Hyvärinen and Oja (2000).

Nonlinear Methods: Autoencoders



- As a non-linear extension of PCA, we can generalize the framework using an autoencoder (Kramer, 1991; Plaut, 2018).
- Autoencoders (AEs) (Bengio et al., 2006) consist of pairs of encoders and decoders that efficiently reduce the dimensionality of data.
- The prefix 'auto' is used because they autonomously learn to encode data in an unsupervised manner.

Images are from Kim et al. (2021).

Nonlinear Methods: Autoencoders

- Specifically, we may consider a flexible encoding function $f_{\text{enc}}(\cdot)$ and a linear decoder W:

$$\boldsymbol{X} \xrightarrow{\text{Embedding}} f_{\text{enc}}(\boldsymbol{X}) \xrightarrow{\text{Reconstruction}} \boldsymbol{b} + W f_{\text{enc}}(\boldsymbol{X}) \tag{78}$$

- To understand its foundation, consider minimizing the reconstruction error over all possible encoding results $f_{\text{enc}}(\boldsymbol{x}_1), \ldots, f_{\text{enc}}(\boldsymbol{x}_n) \in \mathbb{R}^k$ subject to $W^T W = I_k$:

$$n^{-1} \sum_{i=1}^{n} \left\| \boldsymbol{x}_i - (\boldsymbol{b} + W f_{\text{enc}}(\boldsymbol{x}_i)) \right\|^2 \geq \text{tr}(\hat{\Sigma}) - \sum_{j=1}^{k} \widehat{\text{Var}}[\boldsymbol{w}_j^\top \boldsymbol{X}] \tag{79}$$

- It is known that the lower bound is achieved if and only if:

$$\boldsymbol{b} + W f_{\text{enc}}(\boldsymbol{x}_i) = \bar{\boldsymbol{x}} + W^\top (\boldsymbol{x}_i - \bar{\boldsymbol{x}})$$

  This implies that the first k PCs are the optimal choices.
- See Section 14.5.1 in Hastie (2009) for further theoretical details.

Nonlinear Methods: Variational Autoencoders

- Variational autoencoders (VAEs) (Kingma and Welling, 2014) can be understood as a non-linear extension of Independent Component Analysis (ICA) (Khemakhem et al., 2020).
- A more rigorous and detailed discussion on variational autoencoders will be presented in a later chapter.

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